

# Introduction to Statistical Methods

# Contents

|                                              |    |
|----------------------------------------------|----|
| Measures of Central Tendency and Variability | 4  |
| Population, Sample, Parameter, and Statistic | 4  |
| Measures of Center                           | 4  |
| Arithmetic Mean                              | 4  |
| Median, Mode, and Midrange                   | 5  |
| Measures of Variability                      | 5  |
| Five-Number Summary and IQR                  | 5  |
| Worked Example: Suspicious Transactions      | 6  |
| Shape: Symmetry and Skewness                 | 6  |
| Distribution-Based Spread Statements         | 6  |
| Choosing a Visual Summary                    | 6  |
| Axioms and Basic Probability Concepts        | 7  |
| Experiment, Outcome, Sample Space, and Event | 7  |
| Ways to Assign Probability                   | 8  |
| The Probability Axioms                       | 8  |
| Event Relationships                          | 8  |
| Counting Possibilities                       | 9  |
| Product Rule                                 | 9  |
| Permutations and Combinations                | 9  |
| Worked Example: Exactly One Defective Part   | 9  |
| A Reliable Problem-Solving Routine           | 9  |
| Conditional and Total Probability            | 10 |
| Definition                                   | 10 |
| Reading a Contingency Table                  | 10 |
| Independence Through Conditional Probability | 11 |
| The Law of Total Probability                 | 11 |
| Worked Example: Fuel Choice and Full Tanks   | 11 |
| Tables and Trees Express the Same Structure  | 11 |
| Common Errors                                | 11 |
| Bayes Theorem and Naive Bayes                | 12 |
| Bayes Theorem for Two Events                 | 12 |
| Derivation from Conditional Probability      | 13 |
| Worked Example: Defective Cartridge          | 13 |
| From Bayes Theorem to Classification         | 13 |

|                                       |    |
|---------------------------------------|----|
| Worked Naive Bayes Decision . . . . . | 14 |
| Practical Caveats . . . . .           | 14 |
| Final Checklist . . . . .             | 14 |

# Measures of Central Tendency and Variability

Descriptive statistics turn a collection of values into a small set of useful summaries. The goal is not to replace the data with one number. It is to describe where values are centered, how widely they vary, what shape they form, and whether any observations deserve closer attention.

[Detailed](#) | [External Sources](#)

Evidence note: This topic follows contact session 1 in the handout and the substantive content of the 08-08-2026 slide deck. No transcript is available, so the notes do not claim what the professor said beyond the inspectable slides.

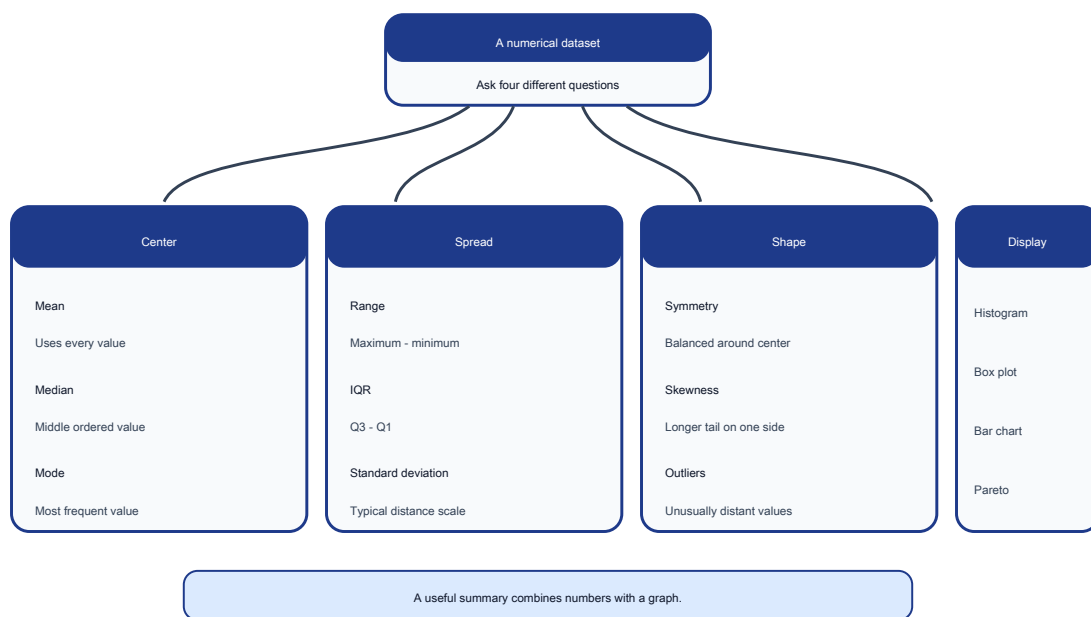


Figure 1. Four complementary views of a numerical dataset. Original course diagram.

## Population, Sample, Parameter, and Statistic

A population is the complete group of interest. A sample is the part that is actually observed. A numerical description of a population is a parameter; a numerical description computed from a sample is a statistic.

For example, the mean salary of every employee in a company is a population parameter, conventionally written  $\mu$ . The mean salary of 50 sampled employees is a sample statistic, conventionally written  $\bar{x}$ . Statistics are used to learn about parameters, and the resulting inference carries uncertainty.

## Measures of Center

### Arithmetic Mean

For values  $x_1, x_2, \dots, x_n$ , the sample mean is:

$$\bar{x} = (1 / n) \sum_{i=1}^n x_i$$

The mean uses every value, so it reacts strongly to extreme observations. A weighted mean is useful when observations have different importance or frequency:

$$\bar{x}_w = (\sum_{i=1}^n w_i x_i) / (\sum_{i=1}^n w_i)$$

Here  $w_i$  is the nonnegative weight attached to  $x_i$ .

## Median, Mode, and Midrange

- The median is the middle value after sorting. With an even number of observations, it is the mean of the two central values.
- The mode is the most frequent value. A dataset can have no mode, one mode, or several modes.
- The midrange is  $(\text{minimum} + \text{maximum}) / 2$ . It is easy to calculate but highly sensitive to extremes.

The median is usually more resistant than the mean when a long tail or an outlier is present. That does not make it universally better: the useful measure depends on the question and distribution.

## Measures of Variability

Center alone can hide important differences. The sets {48, 50, 52} and {10, 50, 90} both have mean 50, but the second has much greater spread.

- Range:  $\text{maximum} - \text{minimum}$ .
- Mean absolute deviation about the mean:  $(1/n) \sum |x_i - \bar{x}|$  for a dataset treated descriptively.
- Population variance:  $\sigma^2 = (1/N) \sum (x_i - \mu)^2$ .
- Population standard deviation:  $\sigma = \sqrt{\sigma^2}$ .
- Sample variance:  $s^2 = (1/(n - 1)) \sum (x_i - \bar{x})^2$ .
- Sample standard deviation:  $s = \sqrt{s^2}$ .

Variance has squared units; standard deviation has the same units as the data. A standard deviation of zero means every value is identical.

Validation note: The slide deck presents division by  $N$ , which is correct when the listed observations are treated as the full population. When data are a sample used to estimate population variance, the conventional unbiased sample variance divides by  $n - 1$ . Keep the interpretation and denominator consistent.

## Five-Number Summary and IQR

The five-number summary contains:

minimum,  $Q_1$ , median ( $Q_2$ ),  $Q_3$ , maximum

The interquartile range is:

$$\text{IQR} = Q_3 - Q_1$$

It measures the spread of the middle 50% of the ordered data. A common box-plot rule flags potential outliers below  $Q_1 - 1.5 \text{ IQR}$  or above  $Q_3 + 1.5 \text{ IQR}$ . The whiskers then extend to the most extreme observed values still inside those fences.

Validation note: Quartile algorithms differ among textbooks and software. State the convention used. The worked example below uses the median-of-halves convention shown in the main slide material.

## Worked Example: Suspicious Transactions

A customer's monthly card transactions, in thousands of rupees, are:

7, 13, 8, 32, 12, 10, 15, 5, 35, 17, 19, 11

- 1 Sort the values: 5, 7, 8, 10, 11, 12, 13, 15, 17, 19, 32, 35.
- 2 The median is  $(12 + 13)/2 = 12.5$ .
- 3 The lower-half median is  $Q1 = (8 + 10)/2 = 9$ .
- 4 The upper-half median is  $Q3 = (17 + 19)/2 = 18$ .
- 5 Therefore  $IQR = 18 - 9 = 9$ .
- 6 The fences are  $9 - 1.5(9) = -4.5$  and  $18 + 1.5(9) = 31.5$ .

The values 32 and 35 exceed the upper fence, so April and September are flagged for investigation. This rule identifies unusual values; it does not prove fraud.

## Shape: Symmetry and Skewness

A distribution is symmetric when its two sides have broadly similar shape around the center. It is right-skewed when the longer tail extends toward larger values, and left-skewed when the longer tail extends toward smaller values.

In common unimodal examples, right skew often gives  $\text{mean} > \text{median}$ , while left skew often gives  $\text{mean} < \text{median}$ . Treat these as patterns, not definitions. A histogram or box plot should accompany numerical summaries.

The slides introduce Pearson's second coefficient of skewness:

$$Sk = 3(\text{mean} - \text{median}) / \text{standard deviation}$$

Its sign describes direction under this convention, but one coefficient cannot fully describe an irregular or multimodal distribution.

## Distribution-Based Spread Statements

For an approximately normal distribution, the empirical rule places about 68%, 95%, and 99.7% of observations within 1, 2, and 3 standard deviations of the mean.

For any distribution with finite mean and positive finite variance, Chebyshev's theorem guarantees that at least

$$1 - (1 / k^2)$$

of values lie strictly within  $k$  standard deviations of the mean for  $k > 1$ . If the variance is zero, every value equals the mean and the distribution is handled directly rather than by standardizing with  $\sigma$ .

Correction: Chebyshev's result is a guaranteed lower bound, not an approximation to the exact percentage. At  $k = 2$ , it guarantees at least 75%; the actual proportion may be larger.

## Choosing a Visual Summary

- Use a histogram for the distribution of one numerical variable. Bin choices affect what is visible.

- Use a box plot to compare center, IQR, and potential outliers across groups.
- Use a bar chart for magnitudes across categories. Bars are separated because categories are not numeric intervals.
- Use a Pareto chart when categories should be ordered from most to least frequent, often with cumulative percentage.
- Use a frequency polygon to connect class-midpoint frequencies and an ogive to show cumulative frequency.

A robust first pass is: inspect the raw values, draw a relevant graph, calculate center and spread, identify unusual observations, and interpret every result in the data's units and context.

For derivations, denominator choices, additional examples, diagnostics, and practice, see [Know More](#).

## Axioms and Basic Probability Concepts

Probability gives a consistent language for uncertainty. It connects an experiment, its possible outcomes, and the events we care about to numbers between 0 and 1. Those numbers become useful only when the experiment and sample space are defined clearly.

[Detailed](#) | [External Sources](#)

Evidence note: This topic maps to contact session 2. The 16-08-2026 slides substantively cover experiments, events, sample spaces, set operations, counting, addition and multiplication rules, mutually exclusive events, and independent events. No transcript is available.

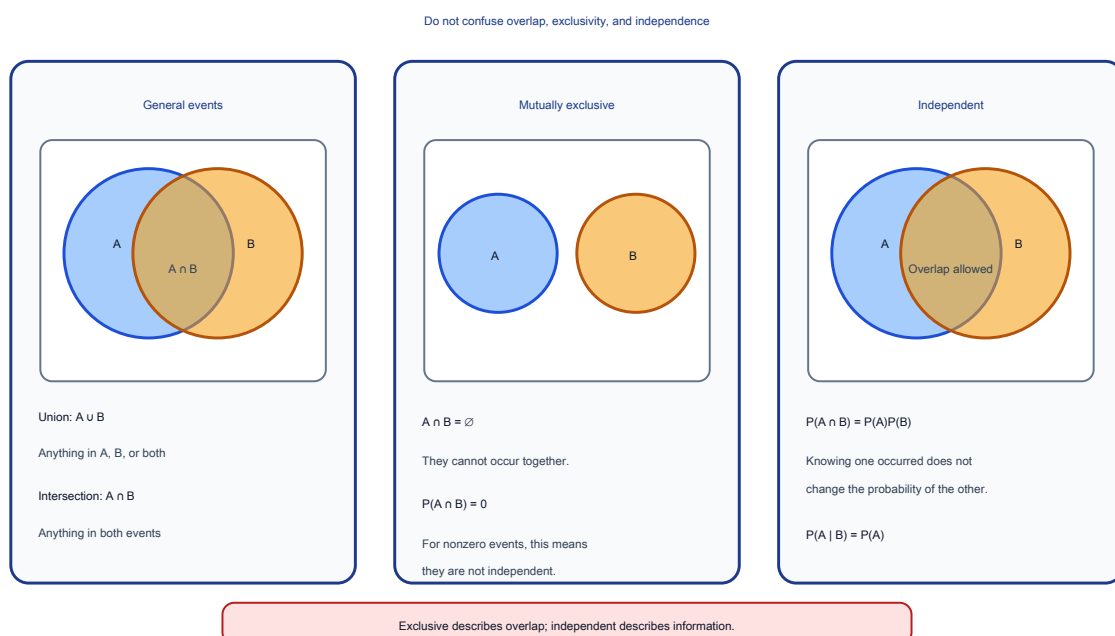


Figure 1. Exclusivity concerns overlap; independence concerns whether information changes probability. Original course diagram.

## Experiment, Outcome, Sample Space, and Event

- A random experiment is a repeatable process whose individual outcome is uncertain.

- An outcome is one possible result.
- The sample space  $S$  is the set of all possible outcomes.
- An event is a subset of  $S$ .
- An elementary event contains exactly one outcome in the chosen sample space.

For one fair die roll,  $S = \{1, 2, 3, 4, 5, 6\}$ . The event “an even result” is  $\{2, 4, 6\}$ ; it is not elementary because it contains three outcomes.  $\{2\}$  is an elementary event.

Correction: The slide deck calls the even-result event elementary. Under the standard discrete definition used here, an elementary event is a singleton outcome, while  $\{2, 4, 6\}$  is a composite event.

## Ways to Assign Probability

The slides introduce three practical viewpoints:

- Classical: When a finite sample space has equally likely outcomes,  $P(A) = |A| / |S|$ .
- Empirical: Relative frequency from repeated observations estimates probability:  $\text{number of observed } A \text{ outcomes} / \text{number of trials}$ .
- Subjective: A probability can encode informed judgment when repeated data or symmetry are unavailable. It should still obey the probability axioms and should be updated when evidence changes.

The classical ratio is invalid if the outcomes are not equally likely. Historical frequency is an estimate from available data, not a guarantee about the next trial.

## The Probability Axioms

For a sample space  $S$  and events in that space:

- 1 Non-negativity:  $P(A) \geq 0$ .
- 2 Normalization:  $P(S) = 1$ .
- 3 Countable additivity: For pairwise mutually exclusive events  $A_1, A_2, \dots$ ,  $P(\cup A_i) = \sum P(A_i)$ .

Useful consequences include:

- $P(\emptyset) = 0$ .
- $P(A^c) = 1 - P(A)$ .
- $0 \leq P(A) \leq 1$ .
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ .

The subtraction in the union formula prevents the overlap from being counted twice.

## Event Relationships

The union  $A \cup B$  contains outcomes in  $A$ , in  $B$ , or in both. The intersection  $A \cap B$  contains outcomes common to both. The complement  $A^c$  contains outcomes in  $S$  but not in  $A$ .

Events are mutually exclusive when  $A \cap B = \emptyset$ , so they cannot occur together. Events are independent when:

$$P(A \cap B) = P(A)P(B)$$

For positive-probability events, this is equivalent to  $P(A | B) = P(A)$ . Mutually exclusive nonzero events are dependent: learning that one occurred makes the other impossible.

## Counting Possibilities

Probability calculations often begin by counting outcomes.

### Product Rule

If one stage has  $m$  choices and a following stage has  $n$  choices, there are  $mn$  ordered combinations. A car with 3 engines, 8 colors, and 4 interior packages has  $3 \times 8 \times 4 = 96$  configurations.

### Permutations and Combinations

For  $n$  distinct objects and  $r$  selections:

- Ordered selection:  $nP_r = n! / (n-r)!$ .
- Unordered selection:  $nC_r = n! / (r!(n-r)!)$ .

With replacement and  $n$  choices at each of  $r$  ordered draws, there are  $n^r$  sequences. Without replacement, available choices decrease after each draw.

## Worked Example: Exactly One Defective Part

A bin contains 2 defective and 4 acceptable parts. Three are selected without replacement.

The total number of unordered samples is:

$$6C_3 = 20$$

To obtain exactly one defective part, select 1 of the 2 defective parts and 2 of the 4 acceptable parts:

$$2C_1 \times 4C_2 = 2 \times 6 = 12$$

Therefore:

$$P(\text{exactly one defective}) = 12 / 20 = 0.60$$

This method works because each three-part subset is equally likely under random sampling without replacement.

## A Reliable Problem-Solving Routine

- 1 State the experiment and sample space.
- 2 Define events in words and symbols.
- 3 Decide whether order and replacement matter.
- 4 Check whether outcomes are equally likely before using a counting ratio.
- 5 Use union, intersection, complement, or conditional structure explicitly.
- 6 Check that the answer lies in  $[0, 1]$  and makes sense in context.

For derivations, counting diagnostics, and additional problems, see [Know More](#).

## Conditional and Total Probability

New information changes the set of possibilities that matters. Conditional probability formalizes that change: instead of measuring an event across the entire sample space, it measures the event inside a stated condition.

[Detailed](#) | [External Sources](#)

Evidence note: This topic maps to contact session 3. The 16-08-2026 slide deck covers conditional probability, multiplication, independence, probability tables, tree-style reasoning, and the total-probability calculation used immediately before Bayes theorem. No transcript is available.

### Definition

For events  $A$  and  $B$  with  $P(B) > 0$ , the probability of  $A$  given  $B$  is:

$$P(A | B) = P(A \cap B) / P(B)$$

The condition  $B$  becomes the new reference set. This is why the denominator is  $P(B)$ , not  $P(S)$ .

Rearranging gives the multiplication rule:

$$P(A \cap B) = P(B)P(A | B) = P(A)P(B | A)$$

Order in the notation matters.  $P(A | B)$  and  $P(B | A)$  usually differ.

### Reading a Contingency Table

The slides use an employment table with 100 male and 55 female employees:

| Position     | Male | Female | Total |
|--------------|------|--------|-------|
| Managerial   | 8    | 3      | 11    |
| Professional | 31   | 13     | 44    |
| Technical    | 52   | 17     | 69    |
| Clerical     | 9    | 22     | 31    |
| Total        | 100  | 55     | 155   |

If  $F$  means female and  $T$  means technical, then conditioning on technical employees restricts the denominator to 69:

$$P(F | T) = 17 / 69 \approx 0.246$$

If the question is reversed, conditioning on female employees gives:

$$P(T | F) = 17 / 55 \approx 0.309$$

The same joint count appears in both calculations, but the denominators differ.

## Independence Through Conditional Probability

Events  $A$  and  $B$  are independent when knowing  $B$  does not change the probability of  $A$ :

$$P(A | B) = P(A)$$

provided  $P(B) > 0$ . Substitution into the multiplication rule gives the equivalent test:

$$P(A \cap B) = P(A)P(B)$$

Do not assume independence merely because events appear unrelated. It is a model claim that should come from the data-generating process or be checked against probabilities.

## The Law of Total Probability

Suppose  $B_1, B_2, \dots, B_k$  form a partition of  $S$ : they are mutually exclusive and together cover the sample space. Then for any event  $A$ :

$$P(A) = \sum_{i=1}^k P(B_i)P(A | B_i)$$

Each term is one disjoint path to  $A$ . Adding the path probabilities gives the marginal probability of  $A$ .

## Worked Example: Fuel Choice and Full Tanks

At a fuel station:

- 40% choose Regular, and 30% of Regular customers fill the tank.
- 35% choose Plus, and 60% of Plus customers fill the tank.
- 25% choose Supreme, and 50% of Supreme customers fill the tank.

Let  $F$  be “fills the tank.” The fuel types form a partition, so:

$$\begin{aligned} P(F) &= P(R)P(F | R) + P(P)P(F | P) + P(S)P(F | S) \\ &= 0.40(0.30) + 0.35(0.60) + 0.25(0.50) \\ &= 0.120 + 0.210 + 0.125 \\ &= 0.455 \end{aligned}$$

Thus 45.5% of customers are expected to fill the tank under the stated model.

The joint probability of choosing Plus and filling is one branch:

$$P(P \cap F) = 0.35(0.60) = 0.21$$

## Tables and Trees Express the Same Structure

A contingency table is convenient when joint counts or proportions are already available. A probability tree is convenient when an experiment unfolds in stages. In a tree:

- multiply probabilities along a branch to obtain a joint probability;
- add mutually exclusive branches to obtain a marginal probability;
- divide a relevant joint probability by the conditioning probability to obtain a conditional probability.

## Common Errors

- Reversing  $P(A | B)$  and  $P(B | A)$ .
- Dividing by the full sample size after a condition has restricted the reference group.
- Multiplying  $P(A)P(B)$  without establishing independence.
- Adding branch probabilities that are not mutually exclusive.
- Forgetting the requirement  $P(B) > 0$  in  $P(A | B)$ .

For a full table derivation, total-probability diagnostics, and practice, see [Know More](#).

## Bayes Theorem and Naive Bayes

Bayes theorem reverses a conditional question. It combines what was believed before an observation with how compatible that observation is under each possible cause. The result is an updated, or posterior, probability.

[Detailed](#) | [External Sources](#)

Evidence note: This topic maps to contact session 4. The 16-08-2026 slides provide a derivation of Bayes theorem, supplier and factory examples, and a categorical naive Bayes classification example. No transcript is available.

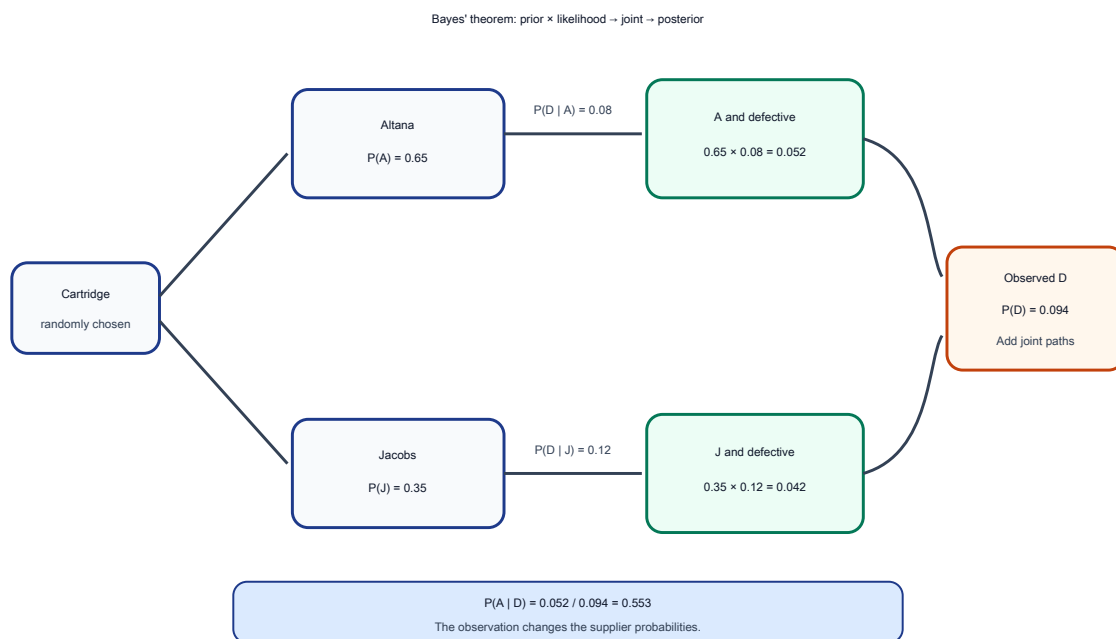


Figure 1. Prior probabilities flow through defect likelihoods to joint and posterior probabilities. Original course diagram.

## Bayes Theorem for Two Events

For events  $A$  and  $B$  with  $P(B) > 0$ :

$$P(A | B) = P(B | A)P(A) / P(B)$$

The terms have distinct roles:

- $P(A)$  is the prior probability of  $A$ .
- $P(B | A)$  is the likelihood of observing  $B$  if  $A$  holds.

- $P(B)$  is the evidence or marginal probability of  $B$ .
- $P(A | B)$  is the posterior probability after observing  $B$ .

Bayes theorem does not say that  $P(A | B) = P(B | A)$ . It shows exactly how the two are related.

## Derivation from Conditional Probability

The multiplication rule writes the same joint event in either conditional direction.

$$P(A \cap B) = P(A)P(B | A)$$

Equivalently,  $P(A \cap B) = P(B)P(A | B)$ . Equating the expressions and dividing by  $P(B)$  gives Bayes theorem.

If  $B_1, \dots, B_k$  partition the sample space, then:

$$\begin{aligned} & P(B_j | A) \\ &= P(B_j)P(A | B_j) / (\sum_{i=1}^k P(B_i)P(A | B_i)) \end{aligned}$$

The denominator is the law of total probability. It adds every mutually exclusive path by which  $A$  can occur.

## Worked Example: Defective Cartridge

Altana supplies 65% of cartridges and Jacobs supplies 35%. Their defect rates are 8% and 12%, respectively. A purchased cartridge is defective. What is the probability that Altana supplied it?

Let  $A$  mean Altana,  $J$  mean Jacobs, and  $D$  mean defective.

$$P(A \cap D) = 0.65(0.08) = 0.052$$

$$P(J \cap D) = 0.35(0.12) = 0.042$$

$$P(D) = 0.052 + 0.042 = 0.094$$

Therefore:

$$P(A | D) = 0.052 / 0.094 \approx 0.553$$

Although Altana supplies 65% of all cartridges, it accounts for only about 55.3% of defective cartridges because Jacobs has the higher defect rate.

## From Bayes Theorem to Classification

Suppose a class variable is  $C$  and a record has features  $X = (x_1, \dots, x_n)$ . Bayes theorem gives:

$$P(C | X) = P(C)P(X | C) / P(X)$$

Naive Bayes assumes the features are conditionally independent given the class:

$$P(X | C) = \prod_{i=1}^n P(x_i | C)$$

For classification:

$$\text{score}(C) = P(C) \prod_{i=1}^n P(x_i | C)$$

Choose the class with the largest score. The common denominator  $P(X)$  is not needed for comparing classes because it is the same for every candidate class.

Validation note: The assumption is conditional independence given the class, not ordinary independence among features across the entire dataset.

## Worked Naive Bayes Decision

The slides classify a customer with:

Age = Young, Income = Medium, Student = Yes, Credit = Fair

The two prior-times-likelihood scores are  $\text{score(Yes)} = (9/14)(2/9)(4/9)(6/9)(6/9) \approx 0.0282$  and  $\text{score(No)} = (5/14)(3/5)(2/5)(1/5)(2/5) \approx 0.00686$ .

Since  $0.0282 > 0.00686$ , the classifier predicts Yes. These are unnormalized scores. Dividing each by their sum would produce posterior probabilities within this two-class model.

## Practical Caveats

- A zero training count can make an entire product zero. Additive smoothing assigns a small nonzero probability to unseen feature values.
- Multiplying many small probabilities can underflow numerically. Implementations usually add log-probabilities instead.
- The largest posterior score may classify well even when the probability estimate is poorly calibrated.
- Correlated features can violate the naive assumption and effectively count similar evidence more than once.

Current-version note: The scikit-learn 1.9 user guide documents additive smoothing for multinomial and categorical naive Bayes and cautions that naive Bayes probability estimates should not automatically be treated as well-calibrated probabilities.

## Final Checklist

- 1 Define the competing causes or classes.
- 2 Verify that their priors form an appropriate partition.
- 3 Use likelihoods in the correct direction.
- 4 Compute all joint paths contributing to the evidence.
- 5 Normalize when an actual posterior probability is required.
- 6 Check whether the result is plausible relative to both the prior and the observed evidence.

For the partition proof, log-score implementation, smoothing, diagnostics, and practice, see [Know More](#).